```
import pandas as pd
df=pd.read_csv("/content/BostonHousing.csv")
df
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lstat	medv
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33	36.2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	9.67	22.4
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	9.08	20.6
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	5.64	23.9
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	6.48	22.0
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396.90	7.88	11.9

506 rows × 14 columns

Next steps: Generate code with df New interactive sheet

```
df.head()
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lstat	medv
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33	36.2

Next steps: Generate code with df New interactive sheet

```
x=df.drop('medv',axis=1)
y=df['medv']
```

```
x.shape
```

```
(506, 13)
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,random_state=0,test_size=0.25)
x_train
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lstat
245	0.19133	22.0	5.86	0	0.431	5.605	70.2	7.9549	7	330	19.1	389.13	18.46
59	0.10328	25.0	5.13	0	0.453	5.927	47.2	6.9320	8	284	19.7	396.90	9.22
276	0.10469	40.0	6.41	1	0.447	7.267	49.0	4.7872	4	254	17.6	389.25	6.05
395	8.71675	0.0	18.10	0	0.693	6.471	98.8	1.7257	24	666	20.2	391.98	17.12
416	10.83420	0.0	18.10	0	0.679	6.782	90.8	1.8195	24	666	20.2	21.57	25.79
...	...	...	...	...	...	...	...	...	...	...	...	...	...
323	0.28392	0.0	7.38	0	0.493	5.708	74.3	4.7211	5	287	19.6	391.13	11.74
192	0.08664	45.0	3.44	0	0.437	7.178	26.3	6.4798	5	398	15.2	390.49	2.87
117	0.15098	0.0	10.01	0	0.547	6.021	82.6	2.7474	6	432	17.8	394.51	10.30
47	0.22927	0.0	6.91	0	0.448	6.030	85.5	5.6894	3	233	17.9	392.74	18.80
172	0.13914	0.0	4.05	0	0.510	5.572	88.5	2.5961	5	296	16.6	396.90	14.69

379 rows × 13 columns

Next steps: Generate code with x_train New interactive sheet

```
x_train.head()
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lstat
245	0.19133	22.0	5.86	0	0.431	5.605	70.2	7.9549	7	330	19.1	389.13	18.46
59	0.10328	25.0	5.13	0	0.453	5.927	47.2	6.9320	8	284	19.7	396.90	9.22
276	0.10469	40.0	6.41	1	0.447	7.267	49.0	4.7872	4	254	17.6	389.25	6.05
395	8.71675	0.0	18.10	0	0.693	6.471	98.8	1.7257	24	666	20.2	391.98	17.12
416	10.83420	0.0	18.10	0	0.679	6.782	90.8	1.8195	24	666	20.2	21.57	25.79

Next steps: Generate code with x_train New interactive sheet

```
x_train.shape
```

```
(379, 13)
```

```
x_test.shape
```

```
(127, 13)
```

```
print('Missing values in x_train:')
print(x_train.isnull().sum())
print('\nMissing values in x_test:')
print(x_test.isnull().sum())
```

```
Missing values in x_train:
crim       0
zn         0
indus      0
chas       0
nox        0
rm         3
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
dtype: int64

Missing values in x_test:
crim       0
zn         0
indus      0
chas       0
nox        0
rm         2
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
dtype: int64
```

```
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy='mean')

# Fit on x_train['rm'] and transform both x_train['rm'] and x_test['rm']
x_train['rm'] = imputer.fit_transform(x_train[['rm']])
x_test['rm'] = imputer.transform(x_test[['rm']])

print('Missing values after imputation:')
print('Missing values in x_train:')
print(x_train.isnull().sum())
print('\nMissing values in x_test:')
print(x_test.isnull().sum())
```

```
Missing values after imputation:
Missing values in x_train:
crim       0
zn         0
indus      0
```

```
chas       0
nox        0
rm         0
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
dtype: int64

Missing values in x_test:
crim       0
zn         0
indus      0
chas       0
nox        0
rm         0
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
dtype: int64
```

```
from sklearn.linear_model import LinearRegression
lr=LinearRegression()
lr.fit(x_train,y_train)
lr.coef_
```

```
array([-1.17662855e-01,  4.41249344e-02, -4.63952593e-03,  2.39658963e+00,
       -1.55520377e+01,  3.74116554e+00, -6.75726349e-03, -1.42527026e+00,
        2.42017434e-01, -1.13810684e-02, -9.87032537e-01,  8.43960264e-03,
       -5.00793354e-01])
```

```
lr.intercept_
```

```
np.float64(37.08471208860728)
```

```
y_pred=lr.predict(x_test)
y_pred
```

```
array([24.93635363, 23.62234668, 29.22051791, 11.98034592, 21.33212014,
       19.48213743, 20.43704478, 21.53736514, 19.00139691, 19.93180273,
        4.9329176 , 16.07457965, 16.93586081,  5.35889713, 39.82294022,
       32.34276014, 22.3382601 , 36.54355707, 31.01504112, 23.32672281,
       24.92147419, 24.27632892, 20.69046113, 30.44558224, 22.467588  ,
        9.94384982, 17.70761282, 17.99175671, 35.64635051, 20.77754315,
       18.1056564 , 17.69701691, 19.75574011, 23.82232226, 29.05416491,
       19.25778587, 10.968351  , 24.57922186, 17.3214642 , 15.1934961 ,
       26.10840915, 20.88492882, 22.25879901, 15.34783337, 22.85347053,
       25.09727164, 19.77254644, 22.71723982,  9.68035541, 24.43316685,
       20.73337829, 17.506121  , 24.46406312, 30.09192253, 13.31142368,
       21.53452495, 20.67704736, 15.36935341, 13.79713445, 22.08266362,
       17.58457714, 21.60690647, 32.90470906, 31.31754118, 17.65624539,
       32.68992319, 18.55902039, 19.09610666, 18.78592905, 23.04260251,
       22.81084125, 23.99335203, 30.64602032, 28.86819717, 25.78537087,
        5.01226167, 36.79263753, 23.80410585, 27.35129557, 19.32617981,
       28.52627919, 19.1700845 , 18.83736974, 37.89486555, 39.2842744 ,
       23.91512333, 25.50524675, 15.70511891, 25.93065697, 16.57119554,
       15.77125589, 12.8759403 , 24.48440767, 30.95204678, 22.36613495,
       20.21105041,  0.24523301, 25.30315525, 15.34881551, 17.84245876,
       25.43785848, 22.44122126, 32.56452175, 22.01909589, 27.37723958,
       23.33826837,  6.3058975 , 14.79829973, 22.43984068, 29.11331423,
       32.96471401, 12.84931899, 19.8206806 , 20.58601504, 12.08971714,
       23.38957098,  4.73332169, 19.82554311,  9.19422275, 44.61520566,
       30.63593066, 12.30508091, 17.5651294 , 21.48739522, 23.637443  ,
       20.29577416, 35.15340559])
```

```
y_pred.shape
```

```
(127,)
```

```
result=pd.DataFrame({'Actual':y_test,'Predicted':y_pred})
result
```

	Actual	Predicted
329	22.6	24.936354
371	50.0	23.622347
219	23.0	29.220518
403	8.3	11.980346
78	21.2	21.332120
...	...	...
49	19.4	17.565129
498	21.2	21.487395
309	20.3	23.637443
124	18.8	20.295774
306	33.4	35.153406

127 rows × 2 columns

Next steps: Generate code with result New interactive sheet

```
residual_error=y_test-y_pred
residual_error
```

	medv
329	-2.336354
371	26.377653
219	-6.220518
403	-3.680346
78	-0.132120
...	...
49	1.834871
498	-0.287395
309	-3.337443
124	-1.495774
306	-1.753406

127 rows × 1 columns

dtype: float64

```
sum(residual_error)/len(residual_error)
```

```
0.4883704614567786
```

```
from sklearn.metrics import mean_absolute_percentage_error
mean_absolute_percentage_error(y_test,y_pred)
```

```
0.17547026405596727
```

```
lr.score(x_test,y_test)
```

```
0.6352134597345098
```

```
from sklearn.metrics import r2_score
r2_score(y_test,y_pred)
```

```
0.6352134597345098
```